COM3031-Probabilistic Machine Learning Workshop: (Bayesian) Linear Regression and Generalised Linear Models

Overview

This workshop consists of two questions. In the first question, I provide a demonstration for the first two parts of Q1. You will complete the remaining parts in the jupyter notebook as provided on ELE. In the second question, you will estimate the parameters of the link function by maximising the log likelihood and estimate the posterior using MCMC. I will publish the solution in python by this week. You may need to go through the lecture slides for Week 2.

Q1

A data set of Scholastic Aptitude Tests (SAT's) and Garde Point Average (GPA) data set has been provided you on ELE. Imagine a linear model between GPA and math SAT score: $y_i = bx_i + \epsilon$, for $i = 1, ..., n$ where $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

1. Estimate b by maximising the likelihood for $\sigma = 0.445$.
2. Estimate the posterior distribution on the parameter b using Markov Chain Monte Carlo (MCMC) for $\sigma = 0.445$. You can assume a normal prior on b.
3. Image you don't know the value of σ, estimate both b and σ by maximising the likelihood.
4. Assign a an appropriate prior on σ. Estimate the posterior distribution for b and σ using MCMC.

Q2

A data set of rain and no-rain (1 and 0 - denoted with $\mathbf{y}$) with humidity values (denoted with $\mathbf{x}$) has been provided to you on ELE. As the data is not normal and has only two outcomes: 1 and 0, a Bernoulli distribution can be used to describe the data:

$$p(y_i|p_i) = p_i^{y_i}(1 - p_i)^{1-y_i}$$

where p_i is the probability of rain on i-th sample.

1. What is the link function (η_i) which can correlate between y_i and x_i linearly?
2. Write the probability p_i in terms of the link function.
3. Use p_i from the previous step and derive the equation of the log likelihood.
4. Maximise the log likelihood on jupyter notebook and estimate the parameters of the link function.
5. Put a normal prior on parameters and estimate the posterior using Markov Chain Monte Carlo.